

Velocity Verlet (APPLICATION)

Monday, 20 July 2026

11:40 PM

$$x_0 = -26072138.45, y_0 = 144774673.82$$

$$v_{x_0} = -29.78893, v_{y_0} = -5.39627 \text{ km/s}$$

$a(t) \rightarrow$ Same from RK4:

$$a_x(t) = 1.0870 \times 10^{-6} \text{ km/s}^2$$

$$a_y(t) = -6.0350 \times 10^{-6} \text{ km/s}^2$$

$$x_{\text{next}} = x_0 + v_{x_0} \cdot \Delta t + \frac{1}{2} a_x(t) \Delta t^2$$

$$= -26072138.45 - 29.78893 \times 3600 + \frac{1}{2} (1.0870 \times 10^{-6}) \times 3600^2$$

$$= -26072138.45 - 107240.15 + 7.04$$

$$= -260179371.56 \text{ km}$$

$$y_{\text{next}} = y_0 + v_{y_0} \Delta t + \frac{1}{2} a_y(t) \Delta t^2$$

$$= 144774673.82 + (-5.39627 \times 3600) + \frac{1}{2} (-6.0350 \times 10^{-6}) \times 3600^2$$

$$= 144774673.82 - 19426.57 - 39.11$$

$$= 144755208.14$$

$$v_{x_{\text{half}}} = v_{x_0} + \frac{1}{2} a_x(t) \Delta t$$

$$= -29.78893 + \frac{1}{2} (1.0870 \times 10^{-6}) 3600$$

$$\approx -29.78677 \text{ km/s}$$

$$v_{y_{\text{half}}} = v_{y_0} + \frac{1}{2} a_y(t) \Delta t$$

$$= -5.39627 + \frac{1}{2} (-6.0350 \times 10^{-6}) 3600$$

$$\approx -5.40713 \text{ km/s}$$

$$\hat{x}_{\text{next}} = 147107006 \text{ km}$$

$$a_{\text{next}_x} = 1.0913 \times 10^{-6} \text{ km/s}^2$$

$$a_{\text{next}_y} = -6.0334 \times 10^{-6} \text{ km/s}^2$$

$$v_{x_{\text{next}}} = v_{x_{\text{half}}} + \frac{1}{2} a_{\text{next}_x} \Delta t$$

$$= -29.78677 + \frac{1}{2} (1.0913 \times 10^{-6}) \times 3600$$

$$\approx -29.78501 \text{ km/s}$$

$$v_{y_{\text{next}}} = v_{y_{\text{half}}} + \frac{1}{2} a_{\text{next}_y} \Delta t$$

$$= -5.40713 + \frac{1}{2} (-6.0334 \times 10^{-6}) 3600$$

$$\approx -5.41799 \text{ km/s}$$

Final position: $(-26179371.56, 144755208.14) \text{ km}$

Final velocity: $(-29.78501, -5.41799) \text{ km/s}$